Certified AC Robust Secure Sets with GENCO in the Loop

Benders bounds · physics-trajectory recalibration of GENCO · conformal screening with a guarantee

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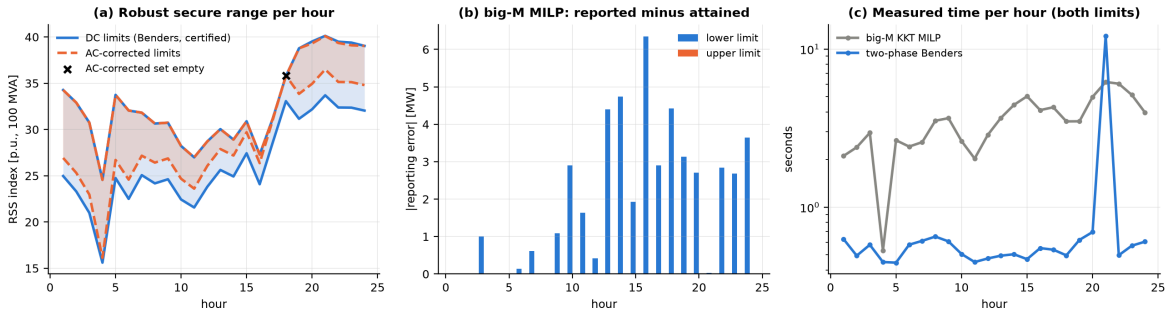
- **Interactive Space:** <SPACE_URL>
- **Code, data recipe, trained weights, results:** <CODE_URL>
- **Paper (IEEE format, 5 pages + 7-page appendix with the complete model and all proofs):** <PAPER_URL>
- **Supplement (complete formulation and proofs, 10 pages):** `paper/supplement.pdf`
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TL;DR

The **robust secure set (RSS)** tells a grid operator, hour by hour, how far the dispatch can move while *every* PV/load realisation in a forecast box stays secure in the base case and in contingencies. In my dissertation it was a DC bilevel problem solved as a big-M KKT MILP. This entry:

1. **Certifies the DC solution** with a two-phase combinatorial Benders decomposition: all 48 hourly problems (IEEE 118-bus, 20 PV buses, two N-1 contingencies) solved to certified optimality. The dissertation's big-M MILP **misreports 20 of 48** problems, by up to **6.3 MW**.
2. **Shows the DC answer is not AC-secure** — 45 of 48 certified DC limits overload a branch in exact AC (worst 114.3 % of rating) — and **fixes it with GENCO in the loop**: GENCO screens worst-case vertices in three topologies, exact AC (gridfm-datakit + lightsim2grid) verifies the flagged ones, and correction cuts go back to Benders. DC **overstates the secure range by 43 %** on average (644 → 414 MW), and in hour 18 the corrected set is **empty** although DC reports 271 MW.
3. **Finds that GENCO knows when it is wrong.** The AC power mismatch GENCO measures inside each of its corrective layers explains **56 %** of its own error. *Physics-trajectory recalibration* (PTR) + split-conformal screening clears **54 %** of secure scenarios without exact AC (plain conformal: 4.7 %), with a **finite-sample guarantee P(insecure ∧ cleared) ≤ α** (α = 5 %, observed 0.09 %) — and **56 %** in a topology GENCO never saw.

Every number in this post is produced by the code (`python genco_ac_rssr_challenge.py`), none is typed by hand.



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1. Why this matters (originality)

GENCO is a neural solver. A security decision needs a *guarantee*. The question this entry answers is **how to use a grid foundation model inside a certified security workflow without giving up the certificate**:

Piece	Role	Guarantee
Two-phase combinatorial Benders	DC robust secure limits	certified bound (master = attained)
Exact AC (gridfm-datakit, lightsim2grid)	verify flagged scenarios, generate correction cuts	exact
GENCO + PTR + conformal	decide which scenarios need exact AC	P(miss) ≤ α, finite sample
GENCO autograd adversary	search the box for AC-worse vertices	heuristic, always verified exactly

New ideas, as far as I know not in the GENCO paper or the RSS literature:

- **Physics-trajectory recalibration (PTR).** GENCO's `GNS_heterogeneous` computes the AC mismatch ‖[ΔP, ΔQ]‖ of its own estimate in each of its K corrective layers. A forward hook records the per-graph trajectory $r^{(1)}…r^{(K)}$ at zero cost. A small gradient-boosted regressor on (GENCO outputs, $r^{(1)}…r^{(K)}$, $\rho = r^{(K)}/r^{(1)}$), fitted on a separate tune split, predicts GENCO's error. No hour, no topology label is used, so it applies to unseen topologies.
- **Conformal screening on top of PTR** with the union-bound guarantee over three security quantities (branch MVA loading, under- and over-voltage). Proof: if a scenario is insecure and cleared, some conformal score exceeds its threshold; each has probability ≤ α/3.
- **Benders without primal big-M** for the RSS bilevel: exact bounds from the data, a robust value-function cut that is tight in 47 of 48 problems, Farkas-based combinatorial cuts, and a dual-lifted phase 2 as fallback.

2. Formulation and guarantees (complete equations and proofs: paper appendix / supplement)

Robust secure set. Nature picks $p = [P_R; P_D]$ in the hourly box $\mathcal{P}_h = [\underline{p}, \overline{p}]$; the operator answers with the dispatch $u \geq 0$. DC line limits in three topologies s (PTDF H_s, derating κ_s), lossless balance and ramp-limited unit bounds form $Ap + Bu \leq b$:

$$-\kappa_s \circ r \le H_s\,(C_G u + C_R P_R - C_D P_D) \le \kappa_s \circ r, \qquad \mathbf{1}^\top u + \mathbf{1}^\top P_R - \mathbf{1}^\top P_D = 0, \qquad \underline{u} \le u \le \overline{u}.$$

With $\mathcal{U}(p) = \{u \ge 0 : Bu \le b - Ap\}$ and the index $\alpha = d^\top u$ (output of the non-balancing units):

$$\overline{\alpha}_h = \min_{p \in \mathcal{P}_h} \max_{u \in \mathcal{U}(p)} d^\top u, \qquad \underline{\alpha}_h = \max_{p \in \mathcal{P}_h} \min_{u \in \mathcal{U}(p)} d^\top u.$$

- **Theorem 1 (meaning of the RSS).** α is achievable for *every* $p \in \mathcal{P}_h$ iff $\alpha \in [\underline{\alpha}_h, \overline{\alpha}_h]$; the set is empty iff $\underline{\alpha}_h > \overline{\alpha}_h$. *Proof:* for fixed p, $\{d^\top u : u \in \mathcal{U}(p)\}$ is the interval $[\psi(p), \varphi(p)]$ (linear image of a polytope); intersect over p.
- **Lemma 1 (vertices).** $\varphi(p) = \max_{\mathcal{U}(p)} d^\top u$ is concave and ψ convex in p, so both worst cases sit at vertices of the box. *Proof:* a convex combination of feasible responses is feasible for the combined p.

Benders master (phase 1) — exact data bounds instead of big-M:

$$u_k \le \overline{u}_k\, z^u_k, \qquad b_i - A_i p - B_i u \le S_i\,(1 - z^\lambda_i), \qquad S_i = b_i - \min_{p,\, 0 \le u \le \overline{u}} (A_i p + B_i u),$$

robust cut $d_2^\top u \ge \varphi_R$ (static LP where only the balancing unit adapts), and combinatorial cuts from Farkas certificates of the dual system $\lambda \ge 0,\ \lambda_{J_\ell} = 0,\ B^\top \lambda \ge d_2,\ (B^\top \lambda)_{J_u} = d_{2,J_u}$:

$$\sum_{k \in C_u} (1 - z^u_k) + \sum_{i \in C_\ell} z^\lambda_i \ \ge\ 1.$$

- **Theorem 2 (certificate).** At every iteration $F(\hat{p}) \le \mathrm{OPT} \le \mathrm{UB}$; phase 1 terminates finitely and is optimal when the dual system is feasible. *Proof:* every bilevel-feasible point with a KKT multiplier λ^* satisfies the master with $z^u = 1[u^* > 0]$, $z^\lambda = 1[\lambda^* > 0]$; a cut excludes only patterns without a dual certificate, and it always excludes the current pattern.
- **Theorem 3 (phase 2).** The dual polyhedron $\{\lambda \ge 0 : B^\top \lambda \ge d_2\}$ does not depend on p, so one constant L^* bounds an optimal vertex multiplier for every p; with $L_d \ge L^*$ the dual-lifted master is exact.

AC correction. Verified overloads tighten $\kappa_{s\ell} \leftarrow \max\big(0.8,\ \kappa_{s\ell}(1-\varepsilon)\,\overline{S}_\ell / S_\ell\big)$.
Proposition 1 (nesting): tighter κ shrinks every $\mathcal{U}(p)$, so the corrected interval lies inside the DC interval, and an empty corrected set stays empty when the cuts of both limits are merged.

PTR + conformal screening. GENCO's per-layer mismatch $\bar{r}^{(l)} = \frac{1}{|\mathcal{N}|}\sum_i \|[\Delta P_i^{(l)}, \Delta Q_i^{(l)}]\|$, $l = 1..K$, and its outputs form z. A regressor g_k fitted on a tune split gives $\tilde{q}_k = \hat{q}_k + g_k(z)$; on a calibration split of size n,

$$c_k = \text{the } \lceil (n+1)(1-\alpha/3) \rceil\text{-th smallest of } q_k - \tilde{q}_k, \qquad \text{clear} \iff \tilde{q}_k + c_k \le \ell_k \ \ \forall k.$$

- **Theorem 4 (no-miss guarantee).** For a scenario exchangeable with the calibration data, $\Pr[\text{insecure} \wedge \text{cleared}] \le \alpha$, for *any* g_k fitted without the calibration data. *Proof:* insecure and cleared implies $q_k - \tilde{q}_k > c_k$ for some k; by exchangeability each such event has probability $\le \alpha/3$ (split-conformal quantile lemma); union bound.

All statements are also checked numerically (`pytest -q tests`: vertex property, concavity, robust bound, nesting, battery monotonicity, conformal quantile lemma).

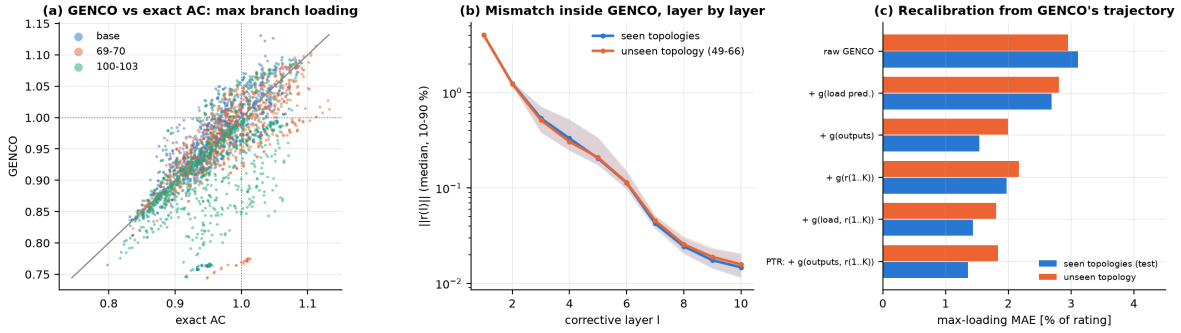
3. Method in one picture

```
hour h -> Benders (DC, certified) -> alpha_DC, worst vertex p*
  pool = {p*, 60 random vertices, GENCO-gradient vertices} x 3 topologies
  GENCO (one batch) -> PTR + conformal -> cleared   (P(miss) <= alpha)
                                       -> flagged   -> exact AC (datakit +
lightsim2grid)
  AC violations -> derate DC branch / slack limits -> Benders again
  stop when no flagged scenario violates
```

4. Results

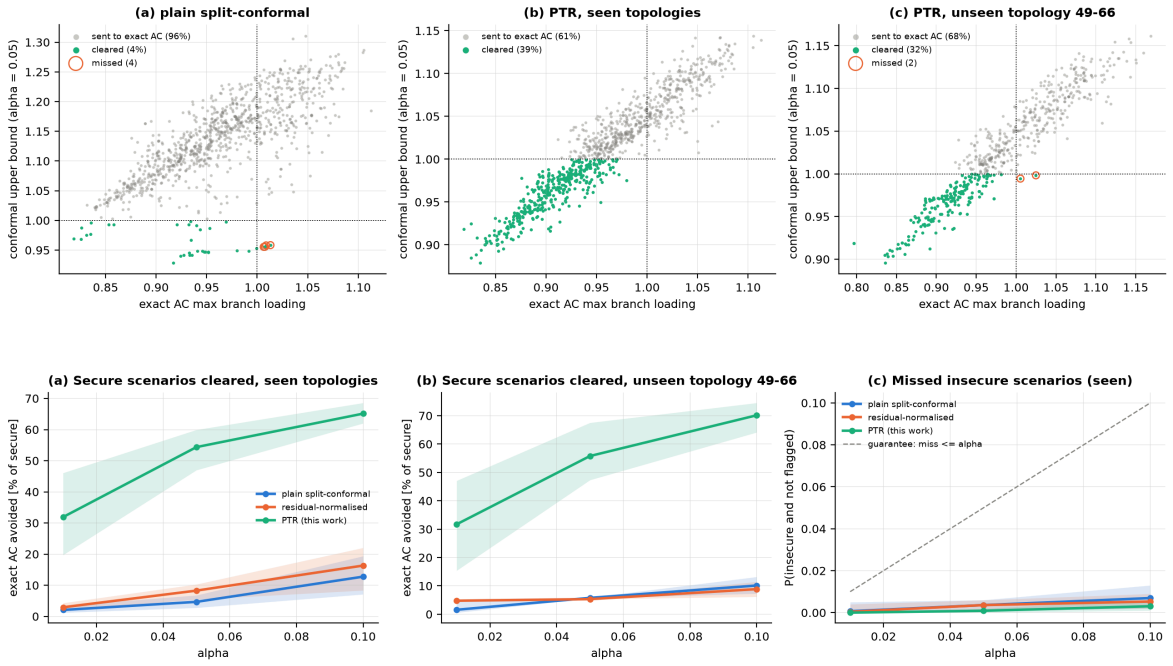
4.1 Certified DC RSS (E1). 48/48 certified optimal; mean 0.51 s (Benders) vs 1.77 s (big-M MILP, same HiGHS solver). The MILP's reported objective differs from what its own leader decision attains in 20 problems (max 6.35 MW, hour 16, lower limit) and in 1 problem its decision is suboptimal — the known big-M pitfall (Pineda & Morales, 2019), now removed.

4.2 GENCO and its corrective-layer trajectory (E3). GENCO (1,077,891 parameters, trained with gridfm-graphkit) predicts the maximum branch loading with MAE 3.14 % of rating on the screening set and 2.95 % in the unseen topology. The mismatch contracts by >100× over the 10 layers, identically in the unseen topology. Single scalars of the trajectory are weak error signals (Spearman 0.23), **the whole trajectory is strong**: it alone removes 37 % of the error, PTR removes 56 %.

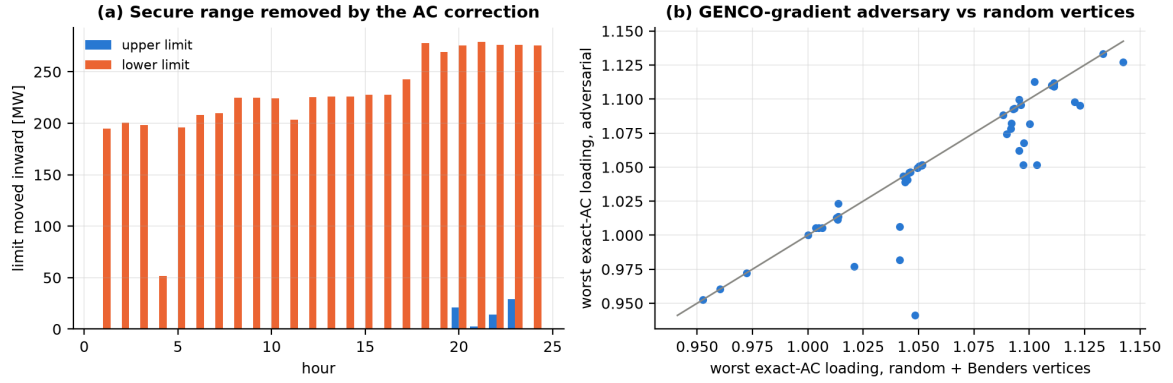


4.3 Screening with a guarantee (E3, 20 random splits). Share of *secure* scenarios cleared without exact AC, and observed miss rate (insecure ∧ cleared):

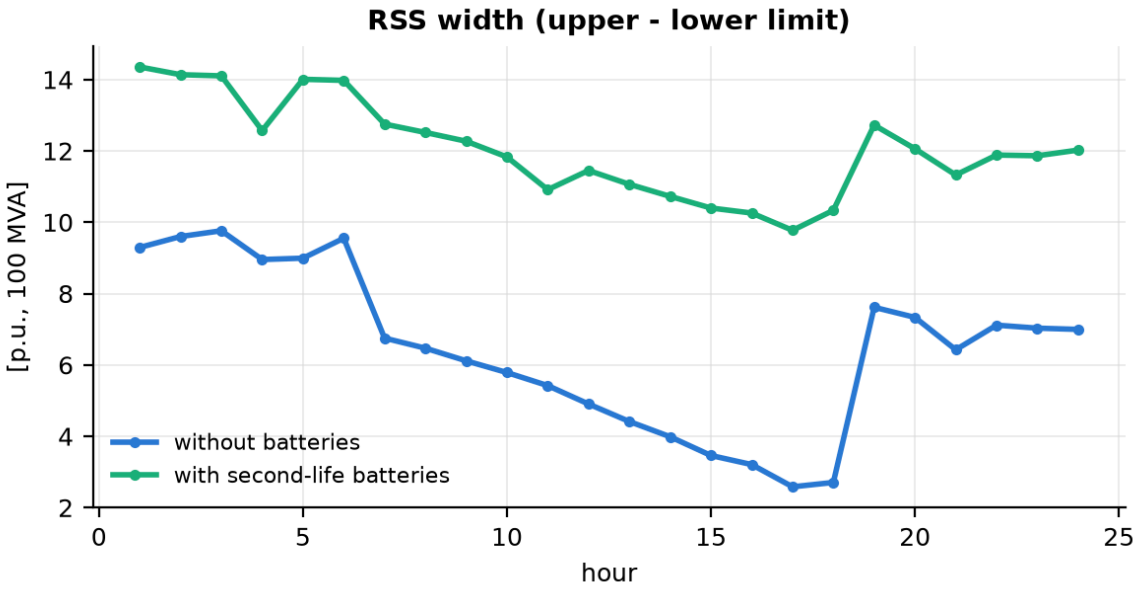
	α = 0.01	α = 0.05	α = 0.10
plain split-conformal	2.1 %	4.7 %	12.8 %
residual-normalised	2.9 %	8.3 %	16.3 %
PTR (this work)	**31.9 %**	**54.4 %**	**65.1 %**
PTR observed miss	0.00 %	0.09 %	0.30 %
PTR, unseen topology 49-66	**31.7 %**	**55.8 %**	**70.2 %**
PTR observed miss, unseen	0.00 %	0.11 %	0.37 %



4.4 AC-corrected RSS (E4, 24 h × 2 limits). 45 of 48 certified DC limits were AC-insecure (worst branch at 114.3 % of rating). After ≤ 3 iterations (29 s per limit) every verified violation is gone; 28 limits moved inward, by up to 279 MW (hour 21, lower); the mean secure range falls from 644 to 414 MW. In hour 18 the corrected lower limit exceeds the upper one: no robust secure dispatch exists in the corrected model, while DC reports 271 MW of range. PTR cleared 35 % of the 19,500 pool scenarios (pools sit on the boundary, so fewer than in E3); a full audit (every pool scenario solved exactly) found **1 missed insecure scenario** (first iteration of one problem; none in any final iteration). The GENCO-gradient adversary beat all Benders and random vertices in 41 of 100 searches, by up to 1.2 % of rating.



4.5 Extension — second-life EV batteries (E5). Batteries at the 20 PV sites (P = SoH × 0.35 × PV capacity, 3.6 p.u. in total) as follower resources widen the certified DC range by 562 MW on average (361–763 MW). The worst-case points are still AC-insecure in 35/48 cases, so the AC loop is needed with batteries too.



5. How GENCO and the gridfm tools are used

- **gridfm-datakit 1.1.0** in power-flow mode with the `lightsim2grid` Newton–Raphson solver generates *all* data: 24,000 training scenarios of the RSS operating region and every exact-AC check. Its Julia OPF step is replaced by our operating-point source (RSS dispatches, PV/load vertices, three topologies); generator set-points come from one AC-OPF per hour.
- **gridfm-graphkit 0.9.0.3**: GENCO (`GNS_heterogeneous`, PowerFlow task, layered physics loss) trained and evaluated through the graphkit CLI/config, normaliser and data module; branch flows via graphkit's `ComputeBranchFlow` (verified against datakit to 10^{-6}).
- **Probe:** forward hook on GENCO's residual layer → $r^{(1)}…r^{(K)}$ per graph, no extra cost.
- **Adversary:** autograd through GENCO w.r.t. PV and load injections, projected sign-gradient ascent of a smooth max of predicted loading over three topologies.

6. Honest limitations

- The official checkpoints (`gridfm/genco-pf-datakit`) could not be downloaded from my compute environment, so a **small GENCO (1,077,891 parameters, 15 epochs, CPU) was trained from scratch**. The code loads the official weights unchanged when available; a stronger model will change (and likely improve) every GENCO number.
- **No speed claim.** On a 118-bus case on CPU, lightsim2grid needs 0.29 ms per scenario and this GENCO 10.3 ms. The value shown here is *how many exact checks can be skipped with a guarantee*, which matters where the exact check is expensive (large grids, AC-OPF re-dispatch, reactive limits, thousands of contingencies, GPU batches).
- The AC-corrected limits are certified for the corrected DC model and verified on the screened pool, not over the continuous box in AC. The correction derates *active-power* branch limits; it reached the 0.8 derating floor in every lower-limit problem (the MVA overload there is partly reactive), so the corrected limits are conservative and the empty set of hour 18 is a statement about the corrected model, not a proof that no

AC-secure dispatch exists. Generator Q-limits are not enforced in the power flow (reported only).
- The conformal guarantee assumes exchangeability; the unseen-topology results are an empirical robustness check, not a guarantee.

7. Reproduce

```
        bash setup.sh                                       # Python 3.12, CPU torch,
gridfm-graphkit, gridfm-datakit
        python genco_ac_rssr_challenge.py                   # data → train → E1 → E3 → E4 →
E5 → figures → numbers
        python genco_ac_rssr_challenge.py --steps e1,e3,e4,e5,fig,numbers,docs   #
reuse the shipped trained GENCO
        pytest -q tests                                     # certificate and conformal-
guarantee checks
        python app.py                                       # Gradio Space locally
```

Thank you to the GridFM team for GENCO, gridfm-graphkit and gridfm-datakit.